Introduction to Exploratory Data Analysis

Course 1. Introduction

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Terminology

Signal
Narrow definition: information than varies in time or space
Broad definition: any type of data, including, for example, attributes of a node in a graph

Data
Narrow definition: tabular, acquired at a given moment in time
Broad definition: any type of data, including, for example, images

Used interchangeably throughout the course.

Terminology

Reconstruction. Usually in the field of signal processing.
Given a set of samples, determine the original *continuous* signal.

Approximation. Traditionally from mathematics.
Given a signal stemming from a complex phenomenon, approximate it using a simpler description.

Estimation. Traditionally from statistics.
Given some data and a model to describe it, determine *the parameters* of the model.

Prediction. Traditionally from statistics.
Given some data, determine *the value* of an instance not present in the dataset.

Aims of Exploratory Data Analysis

Understand the data

- Provide and/or test intuition on the underlying phenomenon: characterize evolution, discover patterns, identify significant events
- Discover biases
- Formulate hypothesis

Prepare the data for approximation/estimation/prediction tasks

- Check data quality
- Remove uncertain/outlying data
- Help choose the right model/algorithm

Present and interpret results

Difference to other types of research

Research involves hypothesis testing.
(Can be one or more hypotheses.)
Usually implies an empirical study and subsequent data analysis that either confirms or rejects the hypothesis.
Can involve multiple iterations.

In exploratory data analysis, there is no explicit hypothesis to test.
There will still be research questions.
Can help formulate hypothesis (which come after).

Understand the data

- Characterize the variables (probability distribution, statistical measures)
- Look for relationships between the variables (correlation, similarity, distance)
- Determine the structure (patterns) in the data (trends, periodicity, peaks, clusters, communities in graphs)

Prepare data for subsequent tasks

- Manage missing data
- Identify outliers
- Preprocess data (encode, scale, subsample, filter)
- Check data linearity, normality etc

Exploratory data analysis

It is called exploratory data *analysis*, but it equally involves:

- **Analysis**. Simplifying, abstracting, dividing/grouping, zooming in/looking into details.
- **Synthesis**. Seeing the whole image, highlighting relations & strucure.

Sources of bias

- **Before data collection**
  Not having a clear methodology or standardized protocols
  - Using different criteria to select data from different groups / cases
  - Not being aware of existing different conditions
  - Not using randomization
  - Using subjective measures with high variability

Sources of bias

- **During the experiments**
  Not applying the same methodology to all experimental cases
  - Observers with different subjective evaluations
  - Delivering prior information to observers that may influence their evaluation (non-blind studies)
  - Interpretation is skewed towards confirm pre-existing beliefs
  - Interpretation is skewed with by new information (recall bias)
  - Collecting data at different times, each having different underlying (seasonal etc) trends
  - Using different methodologies with different temporal implications

Sources of bias

- **During data analysis**
  - Extrapolating beyond the limits of the experiment
  - Comparing to results obtained with different experimental conditions
  - Misinterpreting confounding variables
  - Not including non-significant findings